CS223 Computer Architecture & Organization

Dynamic Scheduling for Multi-Cycle Pipelines

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Dynamic Scheduling

- **Rearrange execution order of instructions to reduce stalls while maintaining data flow**
- Advantages:
  - Compiler doesn’t need to have knowledge of micro-architecture
  - Handles cases where dependencies are unknown at compile time
- Disadvantage:
  - Substantial increase in hardware complexity

How dynamic scheduling works ?

- Limitation of simple pipelining.
  - In-order instruction issue and execution.
  - Instructions are issued in program order.
  - If an instruction is stalled in the pipeline, no later instructions can proceed.
- If instruction j depends on a long-running instruction i, currently in execution in the pipeline, then all instructions after j must be stalled until i is finished and j can execute.

```
DIV.D      F0,F2,F4
ADD.D      F10,F0,F8
SUB.D      F12,F8,F14
```

How dynamic scheduling works ?

- Separate the issue process into two parts:
  - checking for any structural hazards.
  - waiting for the absence of a data hazard.

```
DIV.D F0,F2,F4
ADD.D F10,F0,F8
SUB.D F12,F8,F14
```

- Use in-order instruction issue but we want an instruction to begin execution as soon as its data operands are available.
- out-of-order execution → out-of-order completion.
- OOO execution introduces the possibility of WAR and WAW hazards
- WAR and WAW hazards – solved by register renaming

How dynamic scheduling works ?

- To allow out-of-order execution, **split the ID stage into two**
  - **Issue**—Decode instructions, check for structural hazards.
  - **Read operands**—Wait until no data hazards, then read operands.
- In a dynamically scheduled pipeline, all instructions pass through the **issue stage in order** (in-order issue); however, they can be stalled or bypass each other in the second stage (read operands) and thus enter execution out of order.
- Done by - **Tomasulo’s algorithm**

Register Renaming

DIV.D F0,F2,F4

ADD.D F6,F0,F8

S.D F6,0(R1)

SUB.D F8,F10,F14

MUL.D F6,F10,F8

name dependence with F6

DIV.D F0,F2,F4

ADD.D S,F0,F8

S.D S,0(R1)

SUB.D T,F10,F14

MUL.D F6,F10,T

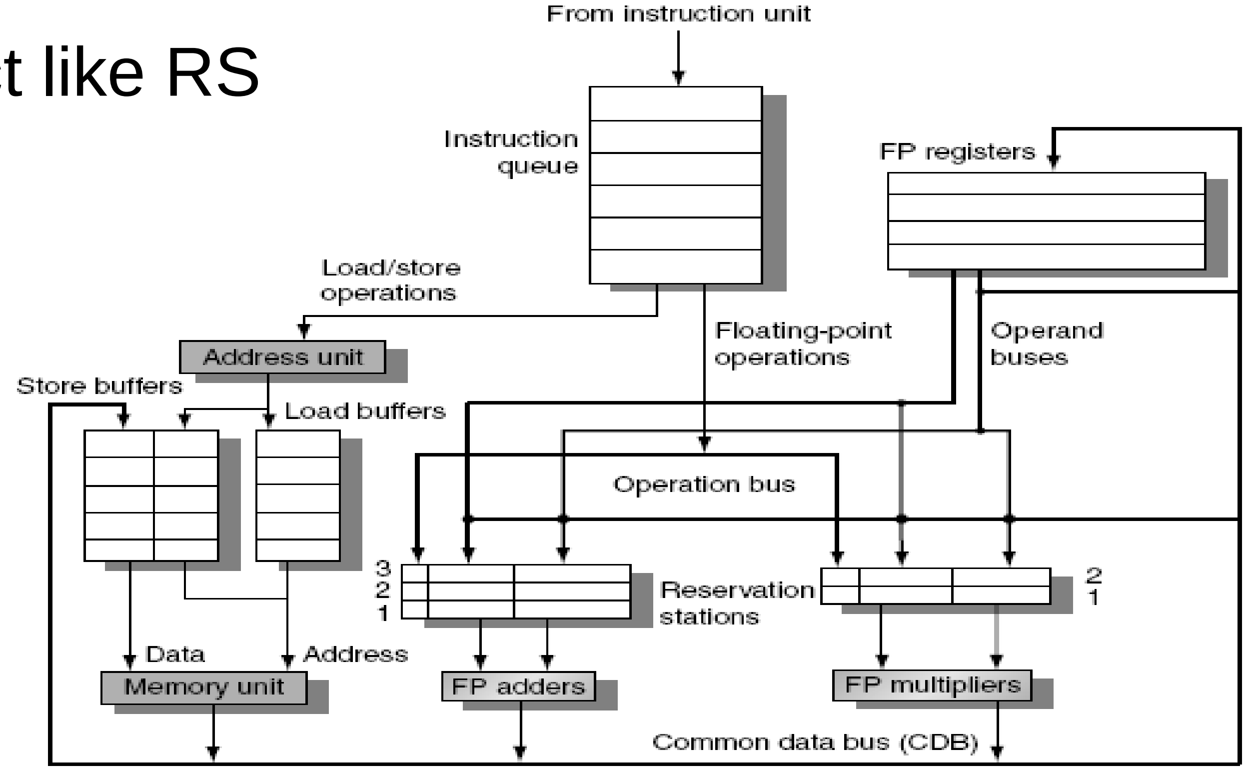
RAW hazard on T

Register Renaming

- Register renaming is done by reservation stations (RS)
- Each RS Contains:
    - The instruction (operation to be done)
    - Buffered operand values (when available)
    - Reservation station number of instruction providing the operand values
- RS fetches and buffers an operand as soon as it becomes available (not necessarily involving register file)
- Pending instructions designate the RS that will provide input
- Result values broadcast on common data bus (CDB)

Tomasulo's Algorithm

- ❖ Load and store buffers contain data and addresses.
- ❖ They also act like RS



Reservation Stations

- **Each reservation station has seven fields.**
- **{Op, Qj, Qk, Vj, Vk, A, Busy}**

1. **Op** —The operation to perform on source operands S1 and S2.

2,3. **Qj, Qk** —The reservation stations that will produce the corresponding source operand; a value of zero indicates that the source operand is already available in Vj or Vk.

4,5. **Vj, Vk** —The value of the source operands.

Only one of the **V field** or the **Q field** is valid for each operand. For loads, the Vk field is used to hold the offset field.

Reservation Stations

- **Each reservation station has seven fields.**
- **{Op, Qj, Qk, Vj, Vk, A, Busy}**

6. **A** —Used to hold information for the memory address calculation for a load or store. Initially, the immediate field of the instruction is stored here; after the address calculation, the effective address is stored here.

7. **Busy** —Indicates that this reservation station and its accompanying functional unit are occupied.

Register Status Indicator

- **The register file has a field, Qi: (RSI)**
- Qi—The number of the reservation station that contains the operation whose result should be stored into this register.
- If the value of Qi = 0 no currently active instruction is computing a result destined for this register, meaning that the value is simply the register contents.

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